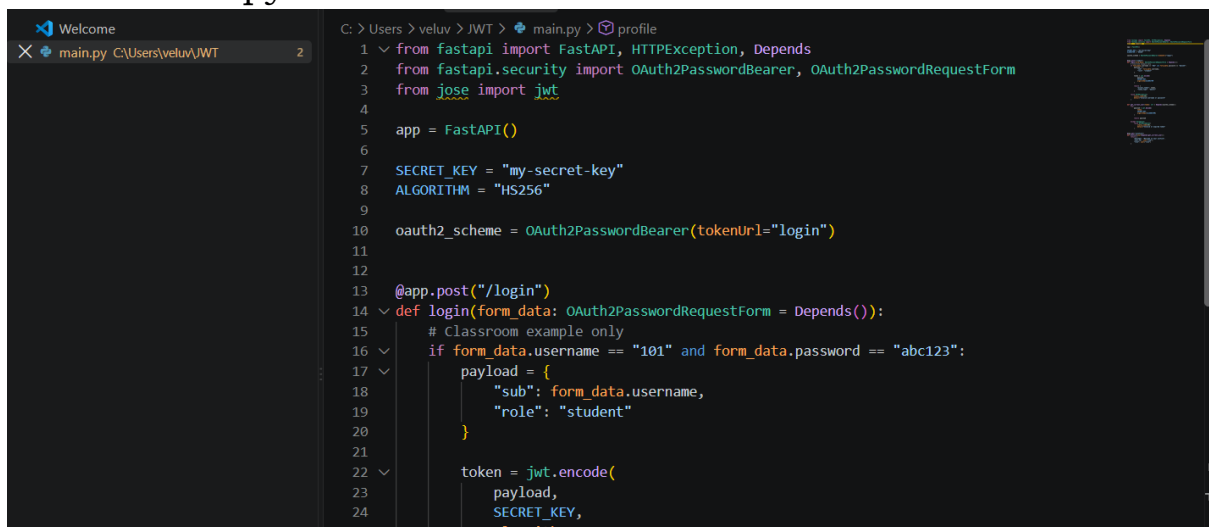


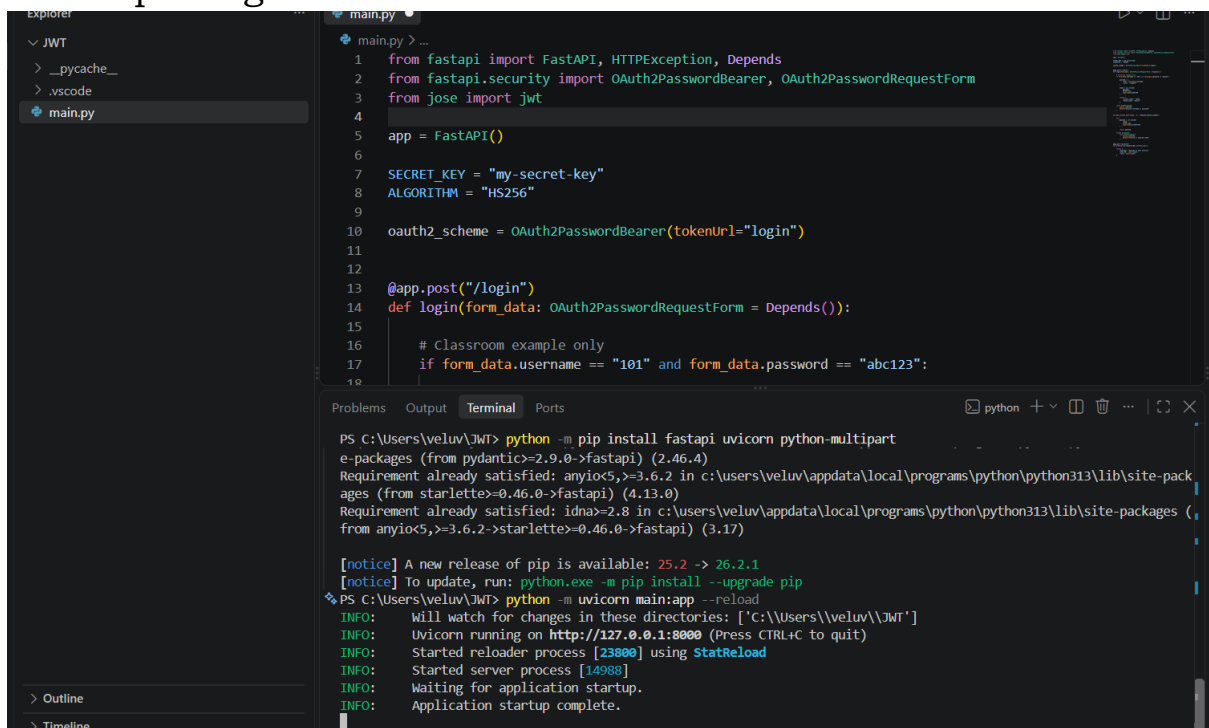
Create main.py



The screenshot shows the VS Code editor with a file named `main.py` open. The file content is as follows:

```
C:\Users\veluv\JWT > main.py
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15     # Classroom example only
16     if form_data.username == "101" and form_data.password == "abc123":
17         payload = {
18             "sub": form_data.username,
19             "role": "student"
20         }
21
22         token = jwt.encode(
23             payload,
24             SECRET_KEY,
25             algorithm=ALGORITHM
```

Install packages

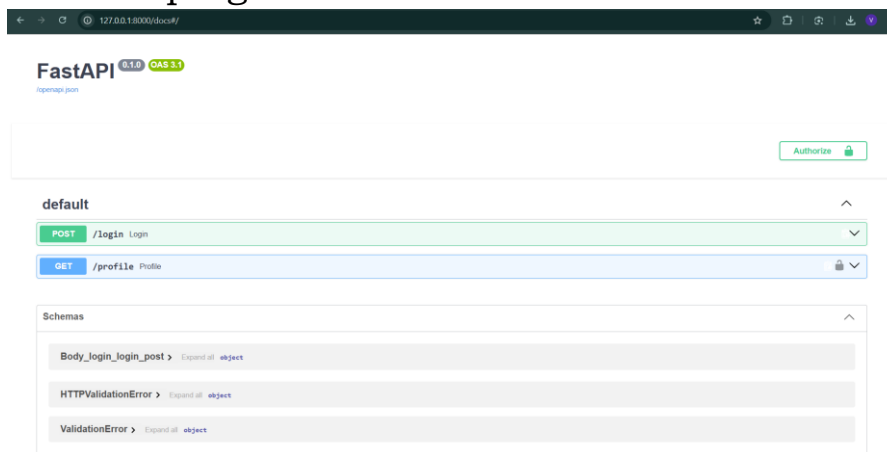


The screenshot shows the VS Code editor with the `main.py` file open. The file content is the same as in the previous screenshot. Below the editor, the Terminal panel shows the following output:

```
PS C:\Users\veluv\JWT> python -m pip install fastapi uvicorn python-multipart
Requirement already satisfied: fastapi in c:\users\veluv\appdata\local\programs\python\python313\lib\site-packages (from pydantic>=2.9.0->fastapi) (2.46.4)
Requirement already satisfied: anyio<5,>=3.6.2 in c:\users\veluv\appdata\local\programs\python\python313\lib\site-packages (from starlette=>0.46.0->fastapi) (4.13.0)
Requirement already satisfied: idna>=2.8 in c:\users\veluv\appdata\local\programs\python\python313\lib\site-packages (from anyio<5,>=3.6.2->starlette=>0.46.0->fastapi) (3.17)

[notice] A new release of pip is available: 25.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\veluv\JWT> python -m uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\Users\veluv\JWT']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [23880] using StatReload
INFO: Started server process [14988]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

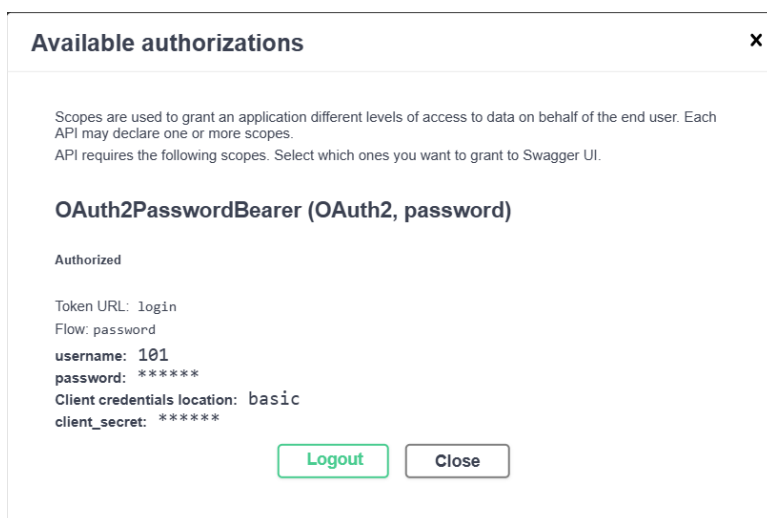
Run the program



Login Test



Authorize JWT



GET /profile response.

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDEiLCJyb2x1Ijoic3R1ZGVudCJ9.NHqkDLPB1Q58FQ1kKmdokvLOjbLUMdH4aRQ6WKzI'
```

Request URL

http://127.0.0.1:8000/profile

Server response

Code	Details
------	---------

200

Response body

```
{
  "message": "Welcome to your profile",
  "user_id": "101",
  "role": "student"
}
```

Response headers

```
content-length: 70
content-type: application/json
date: Tue, 15 Sep 2026 05:37:10 GMT
server: uvicorn
```

Responses